



Data Dictionary

This data dictionary presents the relational database structure shown in the CrayCare Entity-Relationship Diagram (ERD). Each table lists its exact table name and column names, data type, description, and the format or example value used by the current database.

The table and column names are written exactly as defined in the ERD. Primary keys identify each record, while foreign keys connect related records such as users, tanks, batches, sensor data, feeding activities, and notifications.

Table: users

Field Name	Data Type	Description	Format / Example
uid	varchar	Firebase Auth UID. Primary key of the table.	Firebase UID, e.g. {uid}
full_name	varchar	Complete name of the user.	Text
email	varchar	Email address used for sign-in and account recovery.	name@example.com
role	varchar	owner admin.	owner admin
status	varchar	active disabled.	active disabled
tank_id	varchar	owned tank (null for admin). References tanks.tank_id.	Tank document ID, e.g. {tankId}
photo_url	text	Link to the user's profile image.	HTTPS URL
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00

Table: notification_settings

Field Name	Data Type	Description	Format / Example
uid	varchar	one row per user. Primary key of the table. References users.uid.	Firebase UID, e.g. {uid}
sound	boolean	Stores the sound.	true false
vibration	boolean	Stores the vibration.	true false
critical	boolean	Stores the critical.	true false
warning	boolean	Stores the warning.	true false
feeding	boolean	Stores the feeding.	true false
sampling	boolean	Stores the sampling.	true false
operational	boolean	Push preference for automatic actuator and sensor-recovery updates.	true false
updated_at	timestamp	Date and time when the record was last updated.	Timestamp, e.g. 2026-08-31 14:30:00



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Table: hardware_assignments

Field Name	Data Type	Description	Format / Example
id	varchar	always "currentOwner" — single Firestore doc. Primary key of the table.	currentOwner
owner_uid	varchar	Unique identifier of the tank owner. References users.uid.	Firebase UID, e.g. {uid}
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
assigned_by	varchar	admin who assigned. References users.uid.	Admin Firebase UID
assigned_at	timestamp	Date and time when the hardware was assigned.	Timestamp, e.g. 2026-08-31 14:30:00

Table: tanks

Field Name	Data Type	Description	Format / Example
tank_id	varchar	== owner uid by default (1 user = 1 tank). Primary key of the table.	Tank document ID, e.g. {tankId}
owner_uid	varchar	Unique identifier of the tank owner. References users.uid.	Firebase UID, e.g. {uid}
current_batch_id	varchar	nullable. References batches.batch_id.	Text
stocking_date	timestamp	Date when the production batch was stocked.	Timestamp, e.g. 2026-08-31 14:30:00
last_sample_date	timestamp	Stores the last sample date.	Timestamp, e.g. 2026-08-31 14:30:00
sample_count	int	Number of valid readings included for sample.	Whole number, e.g. 20
initial_population	int	Stores the initial population.	Whole number, e.g. 20
initial_total_sample_weight	decimal(10,2)	Stores the initial total sample weight.	Decimal number, e.g. 25.50
initial_total_sample_length	decimal(10,2)	Stores the initial total sample length.	Decimal number, e.g. 25.50
is_initialized	boolean	Shows whether initialized is true or false.	true false
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00

Table: sensor_thresholds

Field Name	Data Type	Description	Format / Example
id	varchar	sensor_name (temperature ph_level dissolved_oxygen turbidity water_level feed_level). Primary key of the table.	Sensor name, e.g. temperature
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}



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Field Name	Data Type	Description	Format / Example
sensor_name	varchar	temperature ph_level dissolved_oxygen turbidity water_level feed_level.	temperature ph_level dissolved_oxygen turbidity water_level feed_level
min_value	decimal(10,2)	Lowest value of the configured normal range.	Decimal number, e.g. 25.50
max_value	decimal(10,2)	Highest value of the configured normal range.	Decimal number, e.g. 25.50
critical_value	decimal(10,2)	Feed level only: critical-below percentage.	Decimal number, e.g. 25.50
hopper_capacity_g rams	decimal(10,2)	Feed level only: calibrated full capacity.	Decimal number, e.g. 25.50
updated_at	timestamp	Date and time when the record was last updated.	Timestamp, e.g. 2026-08-31 14:30:00

Table: sensor_readings

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Document/record ID, e.g. auto-generated ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
temperature	decimal(6,2)	Current water temperature reading.	°C, e.g. 27.20
ph_level	decimal(5,2)	Current pH reading of the water.	Decimal pH, e.g. 7.30
dissolved_oxygen	decimal(5,2)	Current dissolved oxygen reading.	mg/L, e.g. 5.60
turbidity	decimal(8,2)	Current water turbidity reading.	NTU, e.g. 31.00
water_level	decimal(6,2)	Current pond water-level reading.	cm, e.g. 18.00
feed_level	decimal(5,2)	Hopper percentage; operational, not a WQA feature.	0–100%, e.g. 68.00
estimated_feed_g rams	decimal(10,2)	Estimate calculated from level percentage and configured hopper capacity.	Decimal number, e.g. 25.50
recorded_at	timestamp	Date and time when the sensor data was recorded.	Timestamp, e.g. 2026-08-31 14:30:00

Table: sensor_readings_history

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Document/record ID, e.g. auto-generated ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
temp_min	decimal(6,2)	Lowest recorded temp value.	Decimal number, e.g. 25.50
temp_max	decimal(6,2)	Highest recorded temp value.	Decimal number, e.g. 25.50



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Field Name	Data Type	Description	Format / Example
temp_avg	decimal(6,2)	Average recorded temp value.	Decimal number, e.g. 25.50
pH_min	decimal(5,2)	Lowest recorded pH value.	Decimal number, e.g. 25.50
pH_max	decimal(5,2)	Highest recorded pH value.	Decimal number, e.g. 25.50
pH_avg	decimal(5,2)	Average recorded pH value.	Decimal number, e.g. 25.50
DO_min	decimal(5,2)	Lowest recorded dissolved oxygen value.	Decimal number, e.g. 25.50
DO_max	decimal(5,2)	Highest recorded dissolved oxygen value.	Decimal number, e.g. 25.50
DO_avg	decimal(5,2)	Average recorded dissolved oxygen value.	Decimal number, e.g. 25.50
turbidity_min	decimal(8,2)	Lowest recorded turbidity value.	Decimal number, e.g. 25.50
turbidity_max	decimal(8,2)	Highest recorded turbidity value.	Decimal number, e.g. 25.50
turbidity_avg	decimal(8,2)	Average recorded turbidity value.	Decimal number, e.g. 25.50
waterLevel_min	decimal(6,2)	Lowest recorded water level value.	Decimal number, e.g. 25.50
waterLevel_max	decimal(6,2)	Highest recorded water level value.	Decimal number, e.g. 25.50
waterLevel_avg	decimal(6,2)	Average recorded water level value.	Decimal number, e.g. 25.50
feed_level	decimal(5,2)	Operational snapshot; excluded from WQA inputs.	0–100%, e.g. 68.00
estimated_feed_grams	decimal(10,2)	Estimated amount of feed remaining in the hopper.	Decimal number, e.g. 25.50
recorded_at	timestamp	Date and time when the sensor data was recorded.	Timestamp, e.g. 2026-08-31 14:30:00

Table: actuators

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Document/record ID, e.g. auto-generated ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
device_id	varchar	pump aerator1 aerator2.	pump aerator1 aerator2
control_mode	varchar	on off auto (requested by owner).	on off auto
current_state	varchar	on off (actual relay state from ESP32).	on off
last_changed	timestamp	Stores the last changed.	Timestamp, e.g. 2026-08-31 14:30:00



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Table: actuator_logs

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Document/record ID, e.g. auto-generated ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
actuator_type	varchar	pump aerator1 aerator2.	Text
action	varchar	Description of the action that occurred.	Text
type	varchar	on off auto.	on off auto
logged_at	timestamp	Date and time when the event was recorded in the log.	Timestamp, e.g. 2026-08-31 14:30:00

Table: feeder_status

Field Name	Data Type	Description	Format / Example
tank_id	varchar	Unique identifier of the related tank. Primary key of the table. References tanks.tank_id.	Tank document ID, e.g. {tankId}
command_id	varchar	Optional manual request correlation; empty for scheduled feeds.	Command document ID
status_reason	varchar	Human-readable block reason.	Text
status	varchar	idle checking_feed_level dispensing completed skipped_insufficient blocked.	idle checking_feed_level dispensing completed skipped_insufficient blocked
dispense_count	int	completed feeds (Firestore: dispenseCount).	Whole number, e.g. 20
last_seen	timestamp	Firestore: lastSeen.	Timestamp, e.g. 2026-08-31 14:30:00
last_dispensed_at	timestamp	Stores the last dispensed at.	Timestamp, e.g. 2026-08-31 14:30:00
last_dispensed_grams	decimal(8,2)	Stores the last dispensed grams.	Decimal number, e.g. 25.50
feed_level	decimal(5,2)	Estimated percentage of feed remaining in the hopper.	0–100%, e.g. 68.00
estimated_feed_grams	decimal(10,2)	Estimated amount of feed remaining in the hopper.	Decimal number, e.g. 25.50

Table: feeder_schedules

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Schedule document ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}



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Field Name	Data Type	Description	Format / Example
time	varchar	HH:MM.	HH:MM, e.g. 06:00
ampm	varchar	AM PM.	AM PM
time_value	int	minutes since midnight.	0–1439, e.g. 360
grams	decimal(8,2)	Optional; default 20 g. Supported: 20–200 g, multiples of 20. Servo-cycle estimate, not measured weight.	20–200 g; multiples of 20
days	varchar	day-of-week mask "1111111" (Sunday first, 1=active).	7-digit mask, e.g. 1111111
enabled	boolean	Stores the enabled.	true false
is_done	boolean	Shows whether done is true or false.	true false
effective_at_ms	bigint	UTC time stored as milliseconds; creation/edit/re-enable instant.	UTC epoch milliseconds
last_outcome	varchar	completed skipped_insufficient blocked failed; set from device outcome log.	Text
last_occurrence_at	bigint	UTC time stored as milliseconds of the scheduled minute; outcomes are date scoped, no midnight reset.	UTC epoch milliseconds
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00

Table: feeder_logs

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Feeder log document ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
command_id	varchar	Optional originating manual command id; commands are deleted after durable intent.	Command document ID
action	varchar	Description of the action that occurred.	Text
type	varchar	auto manual missed error.	auto manual missed error
status	varchar	completed skipped_insufficient blocked failed.	completed skipped_insufficient blocked failed
schedule_key	varchar	Optional originating schedule document id; not a required FK because schedules may be deleted.	Schedule document ID
schedule_time	varchar	Optional originating schedule time, e.g. 4:00 PM.	Text
occurrence_at	bigint	Original UTC milliseconds since January 1, 1970, retained through offline retries.	UTC epoch milliseconds
requested_grams	decimal(8,2)	Stores the requested grams.	Decimal number, e.g. 25.50
estimated_dispensed_grams	decimal(8,2)	Completed cycles only; not scale-measured mass.	Decimal number, e.g. 25.50
amount_basis	varchar	servo_cycle_estimate.	servo_cycle_estimate



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Field Name	Data Type	Description	Format / Example
estimated_available_grams	decimal(10,2)	Stores the estimated available grams.	Decimal number, e.g. 25.50
feed_level_before	decimal(5,2)	Stores the feed level before.	Decimal number, e.g. 25.50
feed_level_after	decimal(5,2)	Stores the feed level after.	Decimal number, e.g. 25.50
level_change_detected	boolean	Supporting indication only; not proof of exact dispensed mass.	true false
verification_note	varchar	Stores the verification note.	Text
logged_at	timestamp	Date and time when the event was recorded in the log.	Timestamp, e.g. 2026-08-31 14:30:00

Table: feeder_commands

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Command document ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
command_type	varchar	feed_now (manual feeding command).	feed_now
grams	decimal(8,2)	optional; 20.0 default when absent.	20–200 g; multiples of 20
issued_by	varchar	Stores the issued by. References users.uid.	Firebase UID, e.g. {uid}
issued_at	timestamp	Date and time when the command was created.	Timestamp, e.g. 2026-08-31 14:30:00
expires_at	timestamp	App request deadline; ESP also enforces at most 60 seconds after issued_at.	Timestamp, e.g. 2026-08-31 14:30:00

Table: feeder_notification_receipts

Field Name	Data Type	Description	Format / Example
log_id	varchar	tanks/{tankId}/feeder_notification_receipts/{logId}. Primary key of the table.	Feeder log document ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
uid	varchar	Unique identifier of the user account. References users.uid.	Firebase UID, e.g. {uid}
push_attempt_claimed_at	timestamp	Server-only at-most-once claim, not proof of FCM delivery.	Timestamp, e.g. 2026-08-31 14:30:00



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Table: sensor_ingestion

Field Name	Data Type	Description	Format / Example
id	varchar	Logical staging: sensorIngestion/current or current/history/{docId}. Primary key of the table.	current or history/{docId}
hardware_id	varchar	Firestore field: hardwareId.	Text
source_tank_id	varchar	Stores the source tank id.	Text
source_owner_uid	varchar	Stores the source owner uid.	Text
source_assignment_at_ms	bigint	Capture-time assignment identity, not upload-time ownership.	UTC epoch milliseconds
captured_at_ms	bigint	Stores the captured at ms.	UTC epoch milliseconds
routing_status	varchar	quarantined when history cannot be safely assigned.	quarantined
routing_reason	varchar	Stores the routing reason.	Text

Table: machine_learning_assessments

Field Name	Data Type	Description	Format / Example
id	varchar	sortable UTC timestamp doc id. Primary key of the table.	Sortable UTC timestamp ID
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
level	varchar	Good Moderate Poor Critical Insufficient.	Good Moderate Poor Critical Insufficient
model_level	varchar	Raw ML class: Good Moderate Poor Critical.	Good Moderate Poor Critical
rule_level	varchar	Deterministic rolling-risk safety floor.	Text
safety_override	boolean	Stores the safety override.	true false
source	varchar	Model version Rule-based fallback Insufficient data Stale sensor history.	Text
model_algorithm	varchar	RandomForestClassifier Rule-based Not applied.	Text
model_version	varchar	Version embedded in the deployed model artifact.	Text
training_data_origin	varchar	synthetic during prototype; field when retrained from actual pond history.	Text
label_origin	varchar	deterministic_threshold_assessment until independent expert labels are available.	Text
model_feature_count	int	Number of valid readings included for model feature.	Whole number, e.g. 20
analysis_window_minutes	int	Stores the analysis window minutes.	Whole number, e.g. 20
data_status	varchar	ready insufficient stale.	ready insufficient stale
source_recorded_at	timestamp	Latest actual source reading; ISO-8601 in Firestore, nullable.	Timestamp, e.g. 2026-08-31 14:30:00



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Field Name	Data Type	Description	Format / Example
source_age_seconds	int	Stores the source age seconds.	Whole number, e.g. 20
confidence	int	Stores the confidence.	Whole number, e.g. 20
driver	varchar	DO pH temp turbidity waterLevel overall.	DO pH temp turbidity waterLevel overall
driver_label	varchar	Stores the driver label.	Text
driver_value	decimal(10,2)	Stores the driver value.	Decimal number, e.g. 25.50
driver_unit	varchar	Stores the driver unit.	Text
driver_min	decimal(10,2)	Lowest recorded driver value.	Decimal number, e.g. 25.50
driver_max	decimal(10,2)	Highest recorded driver value.	Decimal number, e.g. 25.50
problem	varchar	Stores the problem.	Text
insight	text	Stores the insight.	Text
action	text	Description of the action that occurred.	Text
ts_epoch	bigint	Stores the ts epoch.	UTC epoch time as number
timestamp	timestamp	Stores the timestamp.	Timestamp, e.g. 2026-08-31 14:30:00

Table: batches

Field Name	Data Type	Description	Format / Example
batch_id	varchar	Unique identifier of the related production batch. Primary key of the table.	Batch document ID, e.g. {batchId}
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	Tank document ID, e.g. {tankId}
batch_status	varchar	active harvested superseded.	active harvested superseded
stocking_date	timestamp	wall-clock date/time the batch was stocked. Calendar-day based, NOT a 24-hour rolling clock — setup day = Day 0, next calendar day = Day 1 even if stocked late at night.	Timestamp, e.g. 2026-08-31 14:30:00
harvest_date	timestamp	Date when harvesting was performed.	Timestamp, e.g. 2026-08-31 14:30:00
initial_count	int	Number of valid readings included for initial.	Whole number, e.g. 20
current_count	int	Number of valid readings included for current.	Whole number, e.g. 20
harvest_count	int	Number of valid readings included for harvest.	Whole number, e.g. 20
total_mortality	int	Stores the total mortality.	Whole number, e.g. 20
harvest_weight_grams	decimal(10,2)	Stores the harvest weight grams.	Decimal number, e.g. 25.50
initial_abw	decimal(8,2)	Stores the initial abw.	Decimal number, e.g. 25.50



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Field Name	Data Type	Description	Format / Example
initial_abl	decimal(8,2)	Stores the initial abl.	Decimal number, e.g. 25.50
final_abw	decimal(8,2)	Stores the final abw.	Decimal number, e.g. 25.50
final_abl	decimal(8,2)	Stores the final abl.	Decimal number, e.g. 25.50
days_in_culture	int	Calendar days elapsed since stocking_date (date-only, midnight rollover). Not a 24-hour rolling counter.	Whole number, e.g. 20
sample_count	int	Number of valid readings included for sample.	Whole number, e.g. 20
initial_total_weight	decimal(10,2)	Stores the initial total weight.	Decimal number, e.g. 25.50
initial_total_length	decimal(10,2)	Stores the initial total length.	Decimal number, e.g. 25.50
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00

Table: sampling_records

Field Name	Data Type	Description	Format / Example
id	varchar	Reserved id "baseline" for the initial measurement; weekly samples use auto-id. Primary key of the table.	baseline or auto-generated ID
batch_id	varchar	Unique identifier of the related production batch. References batches.batch_id.	Batch document ID, e.g. {batchid}
sampling_date	timestamp	Date and time when sampling was performed.	Timestamp, e.g. 2026-08-31 14:30:00
avg_body_weight	decimal(8,2)	Stores the avg body weight.	Decimal number, e.g. 25.50
avg_body_length	decimal(8,2)	Stores the avg body length.	Decimal number, e.g. 25.50
sample_size	int	Stores the sample size.	Whole number, e.g. 20
total_weight	decimal(10,2)	Stores the total weight.	Decimal number, e.g. 25.50
total_length	decimal(10,2)	Stores the total length.	Decimal number, e.g. 25.50
biomass	decimal(12,2)	Stores the biomass.	Decimal number, e.g. 25.50
live_count	int	Number of valid readings included for live.	Whole number, e.g. 20
is_baseline	boolean	TRUE = initial measurement at stocking (Day 0); excluded from the weekly cadence and from Week N counting. Weekly samples are is_baseline = FALSE and are due 7 calendar days after stocking_date, then every 7 days from the previous weekly record.	true false
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00



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Table: mortality_records

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Document/record ID, e.g. auto-generated ID
batch_id	varchar	Unique identifier of the related production batch. References batches.batch_id.	Batch document ID, e.g. {batchId}
mortality_date	timestamp	Date when the mortality was discovered or recorded.	Timestamp, e.g. 2026-08-31 14:30:00
mortality_count	int	Number of valid readings included for mortality.	Whole number, e.g. 20
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00

Table: harvest_records

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Document/record ID, e.g. auto-generated ID
batch_id	varchar	Unique identifier of the related production batch. References batches.batch_id.	Batch document ID, e.g. {batchId}
harvest_date	timestamp	Date when harvesting was performed.	Timestamp, e.g. 2026-08-31 14:30:00
harvest_count	int	Number of valid readings included for harvest.	Whole number, e.g. 20
total_weight_kg	decimal(10,3)	Stores the total weight kg.	Decimal number, e.g. 25.50
abw_grams	decimal(8,2)	Stores the abw grams.	Decimal number, e.g. 25.50
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00

Table: notifications

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	Notification document ID
uid	varchar	Unique identifier of the user account. References users.uid.	Firebase UID, e.g. {uid}
notif_type	varchar	critical warning reminder device_auto general.	critical warning reminder device_auto general
title	varchar	Short title shown to the user.	Text
body	text	Complete message shown to the user.	Text
is_read	boolean	Shows whether the user has opened or marked the notification as read.	true false



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Field Name	Data Type	Description	Format / Example
created_at	timestamp	Date and time when the record was created.	Timestamp, e.g. 2026-08-31 14:30:00